[Homework 2: Spring 2025, Due Date: March 10, 2025]

CSE 309: Theory of Automata

Instructions: You are welcome to collaborate on problem sets, provided that (1) you write up your solutions individually, and (2) you clearly cite the names of all collaborators and sources. Failure to do so will result in *zero credit.* An additional key requirement is that you should be able to explain what you submit. Inability to do so will also result again in zero credit. *Direct your all queries to the TA's.*

---

1. (10 points) A string w is said to be a **palindrome** if it reads the same forwards and backwards. Show that the language of all palindromes over the alphabet $\Sigma = \{\texttt{0}, \texttt{1}\}$ is *not regular.*

2. (10 points) Give an *unambiguous* context-free grammar (CFG) for the language A over alphabet $\Sigma = \{\texttt{a}, \texttt{b}\}$:
$$A = \{w : \text{the number of } \texttt{a}\text{'s is at most the number of } \texttt{b}\text{'s in } w\}$$

3. Let $B = \{wuw^{\text{rev}} : u, w \in \{\texttt{0}, \texttt{1}\}^*\}$ (w^{rev} denotes the reverse of w).
   (a) (10 points) Show that B is not regular.
   (b) (10 points) Give a context-free grammar (CFG) for B.

4. (10 points) Design a context-free grammar (CFG) for the context-free language L over the alphabet $\Sigma = \{\texttt{a}, \texttt{b}, \texttt{c}\}$ where:
$$L = \{\texttt{a}^n\texttt{b}^m\texttt{c}^k : k = n + m\}.$$
